DETAILED SPEAKER HANDOUT & PRESENTATION SCRIPT (EN)

Detailed Speaker Handout & Presentation Script (English) — Louis Renner Masterclass

Presenter: Clemens Kunert · Louis Renner GmbH (Meuselwitz Headquarters, Germany)
Conferences: Beijing & Toyohashi Masterclass / Symposium · 33 Main Slides (72 Sub-Sheets) across 5 Chapters
Purpose of this Document: Full stage-ready speaker script for preparation, interpreter briefing, and in-depth lectern reference (complementing the compact 5-page Short Cheat Sheet).

LOUIS RENNER GMBH
Since 1882 · Meuselwitz Headquarters

Stage Controls & Legend: `→` / `Space` advances slides and sub-sheets · `←` steps back · On **Slide 02**, clicking any of the 5 cards jumps straight into that chapter · `G` opens the slide grid · `L` toggles the laser pointer · `▸` marks live interactive elements · `▶` marks embedded videos · `◈` provides natural spoken phrasing · `↪` gives the transition to the next slide.

Chapter 1: Introduction, Heritage & Meuselwitz Plant (Slides 01–04 · approx. 10–12 min)

SLIDE 01 LOUIS RENNER

◈ Spoken Narrative: "Dear colleagues, fellow piano makers, and technicians — a very warm welcome to our Louis Renner Masterclass here in Beijing / Toyohashi. My name is Clemens Kunert from Louis Renner's headquarters in Meuselwitz, Germany. The year 2025/2026 marks a historic milestone for our company: for the first time in more than 140 years since 1882, we have united every single production department — from specialized tonewood milling and grand action manufacturing to complete hammerhead production — under one roof at our modernized headquarters in Meuselwitz."

- **Key Facts on Screen:**
  - **Guiding Theme:** *The Heart of the Piano Since 1882* — over 140 years of action and hammerhead craftsmanship.
  - **Headquarters:** Louis Renner GmbH · Meuselwitz, Germany (at the center of Central Germany's historic piano-building cluster).

↪ Transition: *"Let us first look at Slide 2 to see how today's journey through raw material, precision manufacturing, and concert tone is structured."*

SLIDE 02 Structure and Chapters at a Glance

- **▸ Interaction:** All **5 Bento cards** are clickable and jump directly to their respective chapters (especially handy during Q&A).

◈ Spoken Narrative: "Today we deliberately follow the entire acoustic chain from raw fiber to the finished concert grand on stage: In Chapter 1, we trace our heritage since 1882 and the expansion of our unified Meuselwitz plant. In Chapter 2, we travel just 65 kilometers northwest to Filzfabrik Wurzen to see how pure Merino wool becomes unsanded Weickert hammer felt. Chapter 3 takes us right inside hammerhead production in Meuselwitz. In Chapter 4, I will introduce our ready-to-install Louis Renner Yamaha Kit for C1 to C3 grands, complete with field proof and spectral measurements — and in Chapter 5, we dive deep into the physics and workshop craft of concert voicing."

- **The 5 Chapters at a Glance:**
  1. **Chapter 1 (Slides 3–4):** Heritage since 1882, 14 construction phases & interior drone flight through the unified Meuselwitz plant.
  2. **Chapter 2 (Slides 5–10):** Filzfabrik Wurzen since 1783, Merino wool blend, carding, 4 pillars of fulling, `56%` fulling density & zero sanding of felt sheets.
  3. **Chapter 3 (Slides 11–18):** Climate-controlled felt storage, Weickert Special under SEM, robotic cutting & underfelt, 4 moulding woods & mass scaling, kettle & side jaws, CNC & robotic calibration, riveting & boring.
  4. **Chapter 4 (Slides 19–24):** Louis Renner Yamaha Kit (`C1–C3`) in `4–5 h` on site, India time-lapse, artist verdict & spectral analysis (`H₁–H₁₂` up to `C₈`).
  5. **Chapter 5 (Slides 25–33):** Partial buildup on `A₁`, the hammerhead as a complex spring, Renner Clock Model, 3-string choir phase alignment, string seating, fine & una corda voicing, lacquer dosing/ironing (`max. 135 °C`) & open Q&A.

SLIDE 03 140+ Years of Tradition & Perfection (1882 – Today)

- **▸ Interaction & ▶ Videos:** Step through all **9 historical milestones (`03.1` to `03.9`)** using `→` or `Space`. On **Milestone 8 (2019)** the Steinway clip (`steinwayshort.mp4`) auto-plays; on **Milestone 9 (Today)** the exterior drone video of Meuselwitz (`LouisRennerMeuselwitzAussen.mp4`) auto-plays.

> ◈ **Spoken Narrative:** "Our company chronicle illustrates just how closely precision action mechanics and tonal shaping belong together. Let us walk through the nine defining milestones from Louis Renner's original Stuttgart workshop to Meuselwitz today."

- **All 9 Milestones in Detail:**
  1. **1882 – Founded in Stuttgart (`03.1`):** In October 1882, Louis Renner establishes his workshop on Heusteigstrasse in Stuttgart as a specialized supplier of precision piano actions — laying the foundation of a tradition carried forward in Thuringia since 1990.
  2. **1906 – Launch of In-House Hammerhead Production (`03.2`):** Starting in 1906, Renner manufactures its own hammerheads on custom-engineered presses — matching hammer mass and geometry directly to its own actions. Historical arc from **Bartolomeo Cristofori's** deer-leathered parchment roll (c. 1700) to the tensioned felt hammerhead that became the benchmark for makers such as Steinway & Sons.
  3. **1911 – Individual Electric Drives & Grand Actions (`03.3`):** Building upon **Sébastien Érard's** double-escapement repetition action (1808/1821), **Oscar Renner** launches complete grand piano actions and replaces overhead transmission belts with hundreds of low-vibration individual electric motors — actions and hammerheads now come completely from a single source.
  4. **1944 – Wartime Destruction & Immediate Rebuilding (`03.4`):** During the air raids of **July 24/25, 1944**, the Stuttgart main plant burns to the ground. Yet the know-how lives on in the minds of the seasoned workforce: salvaging usable tooling and rebuilding begins immediately (*"Despite destruction and immense obstacles: Renner will be rebuilt"* — Wilhelm Megenhardt).
  5. **1951–1955 – Dr. Manfred Renner & Global Expansion (`03.5`):** In **1951**, **Dr. Manfred Renner** assumes third-generation leadership, completes the reconstruction of the Stuttgart plant to full pre-war scale by **1955**, and builds international export partnerships with the world's premier piano manufacturers.
  6. **1990 – German Reunification & Zeitz Acquisition (`03.6`):** Following the fall of the Wall in 1990, Renner acquires the experienced action manufacturing plant of the former **Piano Union in Zeitz**. This strategic move retains highly skilled specialists, safeguards major customers such as **Schimmel (`> 40 pianos/day`)**, and establishes our permanent home in Central Germany.
  7. **2011 – Consolidation into Two Modern Sites (`03.7`):** Previously dispersed workshops are consolidated: relocation from Zeitz into a purpose-built `6,800 m²` **production hall in Meuselwitz** (actions, hammerheads, component parts) alongside a **new facility in Gärtringen** (specialized woodworking and administration).
  8. **2019 – Steinway & Sons with Full Operational Autonomy (`03.8` · ▶ Video):** Owner **Clemens von Arnim (`† 2024`)** places Renner into the hands of **Steinway & Sons** rather than short-term financial investors — under one clear condition: **full operational independence**, ensuring Renner continues to supply **all** upright and grand piano makers worldwide openly. *(Brief respectful tribute to Clemens von Arnim).*
  9. **Today – The Unified Meuselwitz Headquarters (`03.9` · ▶ Drone Video):** Launched in autumn 2025, every single department — including specialized woodworking — is now united under one roof at our expanded Meuselwitz headquarters, supported by a new central filtration plant. *(Warm acknowledgment of the long-serving Gärtringen workforce).*

> ↪ **Transition:** *"How did we execute this full-scale plant consolidation without halting production for a single day? Slide 4 documents the transformation step by step."*

SLIDE 04 Construction Documentary: Building the Unified Meuselwitz Plant

- ▸ Interaction & ▶ Video: Advance with → / Space through all **14 construction phases (04.01 to 04.14 , 11.2025 – 09.2026)**; one more step launches 04.15 , the **interior drone flight through the completed factory** (APTA2026-MeuselwitzInnen.mp4).

> ◈ Spoken Narrative: "Between November 2025 and September 2026, we expanded Meuselwitz during live production — with zero delivery downtime for our customers. Keep an eye on three engineering centerpieces: our automated vertical shuttle storage lifts, the 40-ton, 12-meter central filter tower moving 120,000 cubic meters of air per hour, and our rooftop climate plant maintaining 45 to 50 percent relative humidity year-round for wood and felt."

- **Complete Walkthrough of All 14 Phases + Drone Finale:**
  - **Phases 01–03 (Structural Strip-Out & Underground Utilities):** *Phase 1:* Exposing main hall structure for administration & reception · *Phase 2:* Stripping foyer with direct sightline toward felt production · *Phase 3:* Precision underground trenching during uninterrupted factory operation.
  - **Phases 04–05 (Action Line & Smart Vertical Storage):** *Phase 4:* Converting former warehouse bays into our modern piano action / wippen assembly line · *Phase 5:* Installing vertical **shuttle storage lifts** housing tens of thousands of precision parts on minimal floor space.
  - **Phases 06–08 (Heavy Crane Lift of the 40-Ton Filter Tower):** *Phase 6:* Reinforced foundation engineered for 40 tons · *Phase 7:* Millimeter-accurate rigging via **70-ton mobile crane** · *Phase 8:* Mounting the **12 m filter tower** onto a vibration-isolated steel frame.
  - **Phases 09–11 (Hammerhead Move-In & 120,000 m³/h Extraction):** *Phase 9:* Hammerhead department moves into its climate-controlled bay · *Phase 10:* Final assembly of the central filtration system (120,000 m³/h) · *Phase 11:* High-volume overhead extraction ducts linked to every woodworking machine.
  - **Phases 12–14 (45–50% RH Constant Climate & Full Commissioning):** *Phase 12:* Rooftop HVAC and process-air plant · *Phase 13:* Precision humidification (45–50% RH) preventing stress checks in tonewood and dimensional drift in felt · *Phase 14:* Final move-in of the moulding shop — wood, action, and hammerhead production now work side by side under one roof.
  - **Finale (04.15 · ▶ Interior Drone Flight):** Live flight across the bright, dust-free, logistically integrated Meuselwitz production floor.

> ↪ Transition: *"Yet a truly great hammerhead begins even before our workbench — with pure wool and felt. Let us travel 65 kilometers northwest to Wurzen."*

Chapter 2: The Art of Felt Manufacturing – FFW Wurzen (Slides 05–10 · approx. 12–15 min)

SLIDE 05 Filzfabrik Wurzen (FFW) – The Birthplace of Piano Hammer Felt

- ▸ Interaction: Click through the **6 historical era tabs (05.1 to 05.6)** on the left; move the interactive ⌕ Fiber Lens over the macro felt texture on the right.

> ◈ Spoken Narrative: "Filzfabrik Wurzen is not an ordinary supplier — founded in 1783 in Saxony's Mulde valley, it is the oldest industrial felt manufactory in the world. Right here in 1847, deer leather on piano hammers was replaced by German hammerhead felt, and right here the legendary 'Weickert Spezial' formula was born. The fact that Wurzen and our Meuselwitz plant lie just 65 kilometers apart gives us a priceless advantage: our master hammermakers and Wurzen's master feltmakers calibrate every batch together face to face."

- **All 6 Historical Era Tabs at a Glance:**
  1. **1783 – Origin in Wurzen (05.1):** Johann Dankegott Weickert establishes Saxony's wool and felt tradition — specializing early in highly elastic fine felts for piano building.
  2. **1847 – First German Piano Hammer Felt (05.2): Patzschke & Weickert** replace inconsistent leather coverings with homogeneous piano hammer felt (Golzern Mill 1850; aided decisively by the soft, low-mineral water of the Mulde river).
  3. **1850/1851/1867 – World Exhibition Gold (05.3):** Gold medals in Leipzig (1850), London (1851), and Paris (1867) alongside royal visits; leading makers in Vienna, Paris, and London convert to Weickert felt.
  4. **1861 – Industrial Fulling on the Mühlgraben (05.4):** In-house steam power, precision fulling mills, and gentle wool scouring in mineral-poor Mulde water establish reproducible batch consistency.
  5. **1906 – Global Market Leader for 200,000 Pianos/Year (05.5):** Prior to 1914, Director Patzschke and 350 specialists equip up to 200,000 instruments worldwide every year — serving as the benchmark for Steinway & Sons New York.
  6. **Today – Revival of "Weickert Spezial" & Renner Partnership (05.6):** In **2010**, the Brand family revives the authentic 1903 recipe exclusively for concert demands; **just ~65 km distance** to Renner Meuselwitz enables direct master-to-master collaboration without long supply chains.
- **Right Card (4 Acoustic Secrets of Weickert Felt):** 100% pure virgin Merino wool · pure mechanical fulling with zero resins/glues · millisecond elastic rebound of helical keratin springs · 65 km proximity Wurzen ↔ Meuselwitz.

SLIDE 06 Pure Merino Wool & the Secret of Fiber Blending

> **◈ Spoken Narrative:** "Why does Weickert felt sound so alive and stay dimensionally stable across decades? Everything begins with how the raw wool is shipped and scoured. Our overseas Merino wool arrives in Wurzen deliberately in uncompressed bales so the microscopic keratin cuticle scales and strong natural crimp are never crushed. Equally vital is gentle scouring that preserves natural wool grease — lanolin — inside the hair. In Stage 2, the master feltmakers blend long, high-tensile fibers for outer crown tension with fine filler wool for resilient core support."

- **The 2 Process Stages on Screen:**
  - **Stage 01 – Raw Wool Selection:** Loosely packed, **uncompressed overseas Merino wool** (Argentina, South Africa, New Zealand/Australia) · pronounced natural **helical crimp** acting like millions of micro-springs · preserved **natural lanolin** keeping fibers supple for decades.
  - **Stage 02 – Fiber Formula in the FFW Blending Chamber:** Multi-stage gentle plucking and **pneumatic opening** · controlled **climate conditioning** preventing static buildup, fiber breakage, and dense knots · exact balance of **long tensile fibers** (crown) and **fine filler wool** (core cushion).

SLIDE 07 Carding: Parallel Fiber Alignment into Fine Fleece Webs

> **◈ Spoken Narrative:** "After blending, the wool moves onto historic drum cards. Fine rollers gently comb the fibers parallel into a gossamer-thin fleece web and stack them in defined layers. For every piano technician and voicer in this room, right here lies a crucial secret: only because these fibers were combed cleanly parallel during carding does your voicing needle later glide smoothly between the fibers and part them gently, instead of severing tangled fibers!"

- **Key Points to Emphasize:**
  - **Drum Cards (Carding):** Comb millions of Merino fibers without breakage into a defined primary fiber course.
  - **Cross-Layered Stacking:** Builds sheet thickness, tensile strength, and the tapered wedge profile while preventing delamination.
  - **Direct Link to Voicing (Chapter 5):** Parallel fiber alignment = predictable, uniform tonal response to every single needle stitch.

SLIDE 08 The 4 Pillars of Fulling: Steam, Pressure, Motion & Experience

> **◈ Spoken Narrative:** "Now the loose stack of fleece webs enters the hammer fulling mill and becomes true piano hammer felt — driven solely by four physical and artisanal pillars: First, live steam opens the microscopic scales of the wool hair. Second, mechanical pressure compresses the stacked webs. Third, oscillating friction interlocks the barbed scales permanently like burrs. And fourth, the master fuller's trained touch pinpoints the exact second to stop. We use only soft Muldental water and heat — zero synthetic resins, zero chemical binders that would damp the tone."

- **The 4 Bento Cards Exactly as Shown on Screen:**
  1. **1. Live Steam:** Opens microscopic cuticle scales and renders the keratin supple.
  2. **2. Mechanical Pressure:** Progressively compresses the cross-laid fleece webs.
  3. **3. Oscillating Motion:** Rhythmic friction micro-mechanically and permanently interlocks the scales.
  4. **4. Master Experience:** Determines the exact stopping point for maximum natural springback and overtone purity.

SLIDE 09 56% Fulling Density (0.56 g/cm³): The Natural Limit of Mechanical Fulling

> **◈ Spoken Narrative:** "How far can pure Merino wool be compacted purely mechanically? Nature sets the physical ceiling at exactly 56 percent fulling density — 0.56 grams per cubic centimeter. Stop too early, and the hammer lacks internal spring tension. Try to force compaction mechanically beyond 56 percent, and the hollow keratin fibers crush and fracture — what feltmakers call felt breakage, permanently destroying lively rebound."

- **Technical Takeaway:**
  - `56%` (`0.56 g/cm³`) **= Physical Optimum:** Maximum fiber-to-fiber support without structural crushing → shortest string contact times, blooming tone without treble harshness, and lasting density across tens of thousands of strikes.

SLIDE 10 Final Compaction & Tapered 36-Layer Anatomy

- ▸ Interaction: Toggle the animated cross-section on the right between `Unsanded • Shape-Stable` (`10.1`) and `Sanded • Deformation` (`10.2`)!

> ◈ Spoken Narrative: "Slide 10 reveals one of the most decisive quality hallmarks of Wurzen Weickert felt. On the left you see final compaction: after fulling, the sheets are consolidated to 56 to 68 Shore-A using felt hammers and polished natural stones — with zero abrasive sanding! Why is avoiding sanding so vital? Look at the 36-layer cross-section on the right: inside the core, staggered shortened webs form the taper from the slender treble on the left to the thick bass on the right. Enclosing that core on top and bottom, however, run six continuous full-length fleece webs each — 2 times 6 unsanded outer tension cords! Now let's switch to 'Sanded': anyone who sands a felt sheet externally to gauge cuts right through those continuous outer webs. The moment we wrap and press that strip tightly around the wooden hammer moulding, the short inner layers lose their anchor — the crown spreads open and loses its prestress."

- **Comparison of Both Interactive States:**
  - `10.1` **Unsanded (Renner / Weickert):** `2 × 6` continuous outer webs fully enclose the stepped inner webs → closed bending radius around the **hammer moulding**, 100% preservation of outer tensile tautness.
  - `10.2` **Sanded Sheet:** Outer cover webs severed → fibers slip apart under bending tension, causing crown splay and loss of structural prestress.

> ⤷ Transition: *"Now our seasoned, unsanded felt sheet is ready — and travels the 65 kilometers from Wurzen to our factory in Meuselwitz."*

Chapter 3: Hammerhead Production in Meuselwitz (Slides 11–18 · approx. 14–16 min)

SLIDE 11 Climate-Controlled Felt Storage & Assortment Diversity

- ▶ Video: Drone flight through the climate-controlled top-felt sheet inventory in Meuselwitz (`Filzlager_Drone.mp4`).

> ◈ Spoken Narrative: "Welcome back to Meuselwitz! Before a single felt sheet is ever cut, it rests in our climate-controlled inventory. Here we keep more than 30 calibrated FFW and Weickert grades ready at all times — for modern concert grands as well as historic restorations. Regulated humidity, pheromone monitoring, and dedicated resting time allow the final residual fulling stresses to relax completely. Below, you see our underfelt stock in six classic standard colors — from traditional Renner red and wine red through orange, Saxon-Viennese yellow, and classic Bechstein blue to black and custom customer dyeings."

- **Quick Reference Facts:**
  - **Top Felt (`30+ Grades`):** Factory wedge geometry (heavy in the bass, leaner and denser toward the treble) · controlled seasoning & pheromone monitoring.
  - **Underfelt (`6 Standard + Custom Colors`):** Red & Wine Red · Orange · Yellow · Blue · Black · bespoke customer colors.

SLIDE 12 "Weickert Special" – Exclusive to Renner

- ▸ Interaction: Step through the **SEM magnification buttons** on the right (`1× Overview 1 mm` → `10× Fiber Detail 100 µm` → `22× Cuticle Zoom` → optional `⌕ Live Loupe` & `Felt Strip W 1512 K`).

> ◈ Spoken Narrative: "We wanted to go beyond empirical tradition and verify the secret of 'Weickert Special' scientifically in an independent laboratory. So we commissioned the accredited Saxon Textile Research Institute STFI — Test Report P2024 0903. Three numbers tell the entire story: First, 56 percent maximum mechanical fulling density with zero hardeners. Second, a 38 percent higher natural residual wool grease content of 0.36 percent under ISO 3074. And third, the direct consequence for concert life in Asia, Europe, or the Americas: in the 48-hour alternating climate cycling test, Weickert Special swells 65 percent less than the comparison felt — just 4.81 percent instead of 13.69 percent! When we zoom to 22x under the scanning electron microscope on the right, you see why: every single epicuticle scale is completely intact."

- **The 3 Accredited Laboratory Metrics (STFI P2024 0903, M. Kermer):**
  1. `56%` **Max. Fulling Density (`0.56 g/cm³`):** Purely mechanically compacted, zero synthetic resins.
  2. `0.36%` **Residual Wool Grease (`ISO 3074`):** `+38%` more protective natural lanolin vs. `0.26%` comparison sample.
  3. `−65%` **Less Swelling in 48h Climate Test:** Only `4.81%` thickness change vs. `13.69%` → exceptional tuning and voicing stability across seasonal humidity swings (monsoon/summer humidity vs. winter heating).

SLIDE 13 Felt Preparation: Top-Felt Cutting & Underfelt Profiling

- ▶ 2 Simultaneous Factory Videos: 6-axis robot slicing tapered strips at top (`01_roboter_filzzuschnitt.mp4`) · precision underfelt milling at bottom (`02_unterfilz_profilieren.mp4`).

> ◈ Spoken Narrative: “Now the first two manufacturing operations for a set of hammerheads begin — you see both live on the right. At the top, our 6-axis robot slices seasoned felt sheets distortion-free into tapered strips within 1 millimeter tolerance — using a pure cold blade cut with zero fiber crushing or heat glazing, plus inline batch coding on every strip. Below, you see underfelt profiling: the underfelt serves as the elastic coupling bridge between the rigid wooden hammer moulding and the top felt. Its feathered tails are milled thin to plus-or-minus one-tenth of a millimeter so the top felt closes seamlessly around the moulding right up to the highest treble.”

- **The 2 Precision Operations:**
  - **01 · Top-Felt Cutting (`≤ 1 mm Tolerance`):** Distortion-free 6-axis strip cutting · crush-free cold blade cut · full inline batch coding.
  - **02 · Underfelt Profiling (`± 0.1 mm Tolerance`):** Elastic impulse bridge · feathered tails milled to `± 0.1 mm` · symmetrical closing pressure across both flanks.

SLIDE 14 The Hammer Moulding: Tonewoods, Properties & Profiles

- ▸ Interaction: Toggle the right media panel between `Profiles` (technical drawings `Form A, B, US, CSB`) and `Timber Stock` .

> ◈ Spoken Narrative: “The structural backbone of every hammerhead is the wooden hammer moulding. At Renner we machine four selected tonewoods, each with distinct physical characteristics: Hornbeam is a heavy, remarkably stable hardwood — very demanding to season, yet once gently dried the way we do it, an ideal, wonderfully uniform moulding wood. Maple is likewise heavy and often slightly more elastic, though irregular natural sugar deposits carry a higher risk of density variance. Our highest-grade tonewood is American walnut: tight growth rings, long-fibered, and exceptionally light — the supreme wood for a fast, brilliant treble! And Kotibé mahogany is comparatively heavy due to its high natural oil content and is used selectively for specialized applications.”

- **Summary of the 4 Tonewoods & Profile Geometries:**
  - **Hornbeam (`Weißbuche`):** Heavy · demanding to dry · once properly seasoned, ideal & homogeneous.
  - **Maple (`Ahorn`):** Heavy · often slightly more elastic · higher variance risk due to irregular sugar deposits.
  - **American Walnut (`Amerik. Nussbaum`):** Very light · fine-grained & long-fibered · premier choice for a brilliant treble.
  - **Kotibé (`Mahogany`):** Heavy due to high natural oil content · specialized custom applications.
  - **Symmetrical Profiles (`Form B · US · CSB`):** Engineered for Steinway & Sons, Bösendorfer, C. Bechstein, Yamaha, Kawai, Schimmel, Blüthner, Förster, Sauter.
  - **Asymmetrical Profile (`Form A`):** Offset moulding shoulder for vintage Bechstein grands and slender treble scales.

SLIDE 15 Optimal Hammerhead Weight Relative to String Weight

- ▸ Interaction: Step through the zone buttons `① Bass (~15 g)` , `② Break (−1 g)` , `③ Treble (down to 5 g)` ; optionally click any key (A_0–C_8) and test `Too Light` / `Equilibrium` / `Too Heavy` + `Test Strike` to demonstrate contact time τc and string deflection.

> ◈ Spoken Narrative: “Why is low wood mass so critical in the treble, yet governed by completely different physics in the bass? Slide 15 plots both curves on the exact same chart: vibrating string mass and hammerhead weight. In the bass, a single copper-wound string weighs between 90 and over 200 grams — meaning our hammerhead at 15 down to 13 grams is always comparatively light relative to the string! That is why we hold hammerhead weight high as long as possible across the entire bass section to drive maximum fundamental deflection. Moving from Key 23 to Key 24, we cross the break onto plain steel trichords weighing only around 31 grams combined. Because we kept the bass deliberately high, we execute a calibrated one-gram step right here from 13 down to 12 grams! And across the top two octaves up to C8, all three steel wires together drop below 3 grams down to just 0.6 grams, with a half-period shorter than half a millisecond. If the hammer were too heavy here, it would linger on the string and choke its own tone — so we taper smoothly down to 5.0 grams using lightweight American walnut.”

- **The 3 Register Scaling Laws:**
  1. **① Bass (`Keys 1–23` , copper-wound `90–210 g`):** Hold hammer mass (`~15.0 → 13.0 g`) high as long as possible for deep fundamental power.
  2. **② Break (`Key 23 → 24` , plain steel `3 strings ≈ 31 g`):** Calibrated `−1.0 g` **step** (`13.0 → 12.0 g`) for a seamless register transition.
  3. **③ Treble (`last 2 octaves to C8`):** Choir mass drops from `2.7 g` (C_6) to `~0.6 g` (C_8 , `~0.2 g` /wire, $\frac{1}{2}$ `T < 0.5 ms`) → smooth taper to `5.0 g` for instantaneous rebound without self-damping.

SLIDE 16 Pressing & Gluing to 0.1 mm

- ▶ **Factory Video:** Precision glue-press loading and closing in Meuselwitz (`04_hammerkopf_pressen_leimen.mp4`).

> ◈ **Spoken Narrative:** "Now we unite top felt, underfelt, and hammer moulding in our custom glue presses to one-tenth of a millimeter accuracy. Watch the two-stage kinematics closely in the video: First, the tapered felt strip is driven vertically down into the kettle — building immense circumferential tension across those unsanded outer fleece webs. Second, the heated side jaws close inward from left and right, pressing the felt tails symmetrically onto the glue-coated faces of the hammer moulding. Right in that instant our complex tension spring is born: maximum tensile tautness outside over a densely compressed, resilient cushion inside."

- **Key Terminology (EN / DE):**
  - **Kettle** (`Pressgesenk`) → shapes the crown and stretches the outer skin.
  - **Heated Side Jaws** (`temperierte Gegenlagen`) → close to **`0.1 mm` tolerance** and lock the tension permanently onto the glue joint.

SLIDE 17 CNC Milling & Robotic Calibration

- ▶ **2 Sequential Factory Videos:** Video 1 shows the CNC milling center (`CNC_Bearbeitung.mp4`, `17.1`), then automatically cross-fades to Video 2: single-head calibration on the robotic cell (`GLT_Robot_Sanding_HK_Renner.mp4`, `17.2`).

> ◈ **Spoken Narrative:** "Turning a pressed strip into 88 perfectly matched individual hammerheads combines 5-axis CNC machining and robotics across three stages: First, our CNC centers profile the raw wooden hammer mouldings to 0.1 millimeter precision. Second, after pressing and curing, the side felt overhang is cleanly finish-milled on the CNC unit. And third — as you now see in the second video — our robotic cell picks up each sliced individual hammerhead and calibrates its surface and contour with total repeatability."

- **The 3 Stages in Exact Order:**
  1. **CNC-Machine the Hammer Mouldings (`0.1 mm` Tolerance):** Exact wood geometry prior to pressing.
  2. **Trim & Finish-Mill After Pressing & Gluing:** Clean removal of lateral felt overhang along the moulding contour.
  3. **Single-Head Robotic Calibration:** Reproducible surface and contour calibration of every individual hammerhead.

SLIDE 18 Automated Riveting & Boring

- ▶ **Factory Video:** Automated shank boring and brass rivet insertion (`05_hammerkopf_bohren_nieten.mp4`).

> ◈ **Spoken Narrative:** "In the final production step, every individual hammerhead receives its model-specific action geometry: each note from 1 to 88 is bored at its exact individual boring angle, exact depth, and precise shank diameter — 5.4 millimeters in this example. Simultaneously, the machine drives a brass rivet through the glued felt tails so the immense internal tension can never creep open even after decades of concert play, and contours the hammer tail radius for clean, reliable checking against the backcheck felt."

- **The 3 Quality Hallmarks:**
  1. **Precision Boring Angle:** Individual angle, depth, and bore diameter guidance (e.g., `ø 5.4 mm`) note by note.
  2. **Brass Riveting:** Permanent mechanical reinforcement of the bonded felt tails.
  3. **Radiused Hammer Tail:** Cleanly contoured tail radii for dependable checking.

> ↪ **Transition:** *"Traditionally, this is where days of workshop labor begin for the piano technician — boring, shank gluing, travelling, and rough pre-voicing. Over five years of R&D, we engineered a breakthrough solution for one of the world's most common grands: our Louis Renner Yamaha Kit."*

Chapter 4: The Louis Renner Yamaha Kit C1–C3 (Slides 19–24 · approx. 12–15 min)

SLIDE 19 Louis Renner Yamaha Kit (C1–C3) – Seamless Drop-In

- ▸ Interaction: Walk through all **5 topic tabs (`19.1` to `19.5` : `SYSTEM` · `TOUCH` · `DROP-IN` · `TONE` · `TIME SAVED`)** on the left; use the `⌕ Macro Zoom` lens over head, shank, and flange on the right.

> ◈ Spoken Narrative: "Tens of thousands of Yamaha C1, C2, and C3 grands stand in conservatories, recording studios, recital halls, and homes worldwide — mechanically superb instruments where pianists frequently wish for a warmer, singing European concert tone, especially once the original hammers have packed down and hardened. Over five years, we developed a proprietary measurement and production system that calibrates these hammerheads so precisely to the acoustic target of the C1 to C3 scale that the set delivers a full, warm concert sound straight out of the box — with zero manual pre-voicing required! That gives complete confidence even to technicians who do not voice concert grands every single day. Better yet: the heads come pre-bored at exact angles and bonded with rock-hard natural hide glue onto quarter-sawn hornbeam shanks with pre-tensioned knuckles and matching flanges. You fasten a single screw per note straight onto the original Yamaha rail — completing the entire conversion on site in four to five hours instead of hauling the piano into the workshop for three to seven days!"

- **All 5 Tabs for Presentation:**
  1. **Tab 1 – `SYSTEM` (`19.1`):** 5 years of R&D · proprietary measurement & production system · immediate concert tone **with zero manual pre-voicing** (surpassing even factory-fresh original sets).
  2. **Tab 2 – `TOUCH` (`19.2`):** Lightweight calibrated **walnut moulding** + stiff quarter-sawn **hornbeam shanks** + knuckles glued under tension → reduced moment of inertia, noticeably lighter and faster repetition.
  3. **Tab 3 – `DROP-IN` (`19.3`):** Pre-bored at exact angles and factory-glued onto shanks & flanges with traditional rock-hard **natural hide glue** → **1-screw mounting** directly onto the original Yamaha aluminum rail (`0 h` boring labor, zero drilling risk).
  4. **Tab 4 – `TONE` (`19.4`):** Pressed from **one single homogeneous Weickert felt strip** → seamless note-to-note continuity across all 88 keys, articulate bass, bell-clear cantabile treble.
  5. **Tab 5 – `TIME SAVED` (`19.5`): `4–5 hours` on site** instead of `3–7 workshop days` (no piano transport, no extracting/boring/hot-gluing) → full focus on fine regulation and room-specific tonal finishing!

SLIDE 20 Time-Lapse Installation: Complete Conversion in One Day

- ▶ Video: On-site time-lapse documentary filmed in India (`Renner_Timelapse_-_Timestamp.mp4` , `10:00 – 18:00`).

> ◈ Spoken Narrative: "To prove this is real-world workshop practice rather than brochure theory, watch this unedited time-lapse filmed on location in India: At 10:00 a.m., conversion begins right where the grand stands — removing the worn original set, fastening our pre-assembled kit with one screw per flange onto the original aluminum rail, aligning to the strings, performing complete fine regulation of blow distance, let-off, and checking, plus tuning. By 6:00 p.m. that very afternoon, the grand stands concert-ready for the pianist."

- **Workflow Chain on Screen:** `10:00–18:00` (1 workday on site) · Removal → Plug-&-Play 1-screw mounting → Fine regulation (blow distance, let-off, checking) → Tuning → Concert test · **Zero transport risk**.

SLIDE 21 Pianist Feedback: Pianist Raaga in Action

- ▶ Video with Audio: Immediate artist feedback recorded with Pianist Raaga right after conversion (`Raaga_Speech.mp4` — **unmute room audio!**).

> ◈ Spoken Narrative: "Let us listen directly to Pianist Raaga's immediate impression right after playing the first phrases on the converted instrument — both tonally from whispered pianissimo to fortissimo and mechanically under her fingers." (Let video play, then briefly recap:) "That exact blend of European warmth, singing melodic projection, and effortless dynamic control is what technicians report on every installation."

SLIDE 22 Sound Spectrum Comparison: Original vs. Renner Yamaha Kit

- ▸ **Interaction:** Click across the 88-key keyboard or register buttons to inspect **2-octave windows**; toggle `Original`, `Yamaha Kit`, `Gain`, and `H1–H8`.

> ◈ **Spoken Narrative:** "As engineers and piano makers, we wanted to quantify that subjective musical impression objectively in decibels. So we analyzed a nearly twelve-minute chromatic and musical reference recording captured before and after conversion under identical microphone placement: 0:00 to 5:17 with the original hammers, and 5:18 to 11:42 with the Renner Yamaha Kit. Look at the power spectral density curve: in the low bass, muddy boominess drops by more than 3 decibels; the warm fundamental body from 250 to 800 Hertz stays 100 percent intact; and right where a grand must sing and project in a hall — from 800 to 6,000 Hertz — the Renner Kit unlocks massive harmonic energy!"

- **All 5 Measured Spectral Bands for Reference:**
  1. **1. Bass (`50–250 Hz`):** `−3.28 dB` (`−53%` mud/boom) → crystal-clear bass chord separation.
  2. **2. Fundamental Body (`250–800 Hz`):** `−0.07 dB` (`100%` intact) → full foundational warmth preserved.
  3. **3. Singing Presence (`800–2,500 Hz`):** `+2.53 dB` (`+79%` acoustic energy; peaking at `+3.4 dB` **at** `1.45 kHz`) → carrying concert tone.
  4. **4. Brilliance (`2.5–6.0 kHz`):** `+1.87 dB` (`+54%` energy) → bell-clear overtones from Weickert felt.
  5. **5. Air (`6.0–12.0 kHz`):** `+0.31 dB` (`+7%`) → crisp attack definition with zero metallic glare.

SLIDE 23 Partial Distribution (H_1–H_{12}) & 24-Semitone Comparison

- ▸ **Interaction:** Switch register windows (`A0–A2 Bass` through `A5–A7 Brilliance`) and display modes (`Relative H₁` · `Δ Gain` · `Absolute dB`).

> ◈ **Spoken Narrative:** "Slide 23 decomposes every note into its first twelve harmonic partials H1 through H12. On the left you see the acoustic recipe of noble piano tone: the even partials — octave H2, double octave H4, H6, and H8 — give the sound warmth, body, and richness. The odd partials — fifth H3, major third H5, and H7 — create singing cantabile core and hall projection. And on the right, across 24 consecutive semitones, you see proof of our single-strip felt homogeneity: overtone gain never jumps erratically from note to note, but flows smoothly across the entire keyboard."

SLIDE 24 4 Tonal Gains & Overtone Retention up to C_8

- ▸ **Interaction:** Drag the **pitch gain calculator (`up to C₈ = 4,300 Hz`)** on the right (e.g., to `D♯₇ = 2,500 Hz` or `C₈ = 4,186 Hz`) to read live net dB gain.

> ◈ **Spoken Narrative:** "Slide 24 summarizes our four measured acoustic gains: First, plus 2.53 decibels — 79 percent more sound energy — in the singing register. Second, plus 1.87 decibels in the brilliance band. Third, minus 3.28 decibels less low-end boominess. And fourth, look at the spectral roll-off slope across all dynamic levels from pianissimo to fortissimo on the right: whereas the original set drops off at minus 7.38 decibels per kilohertz, the Renner Kit sustains upper harmonics at minus 6.45 decibels per kilohertz — a 12.6 percent flatter slope right up to C8 at 4,186 Hertz!"

> ↪ **Transition:** *"Having seen how an optimally calibrated hammerhead performs straight from the factory, let us open the master chest of concert voicing in Chapter 5: how do we sculpt tone deliberately with needle, file, iron, and mass?"*

Chapter 5: The Art of Concert Voicing & Finale (Slides 25–33 · approx. 18–20 min)

SLIDE 25 Partial Structure on A_1 (55 Hz): Felt Hardness & Strike Point

- ▸ Interaction: First toggle `☁ Soft`, `✦ Concert Ideal`, and `⚡ Hard` at top + click `Play A₁ (55 Hz)`; then drag `2. Strike Point (x₀ / L)` across `1/9 L`, `1/8 L`, and `1/7 L`.

> ◈ Spoken Narrative: "Every piano note is an architecture of harmonic partials — let us examine low A1 at 55 Hertz with a speaking string length of 1,560 millimeters. As piano makers, we command two fundamental physical levers: The first lever is crown hardness and thus hammer-string contact time Tau-c. When the crown is voiced softer, the hammer lingers on the string for roughly 2.85 milliseconds — acting as a low-pass filter that lowers upper partials H6 through H12 by 4.6 decibels for a dark, rounded tone. When the crown is too hard, it rebounds in just 1.25 milliseconds, driving upper partials up by nearly 5 to over 6 decibels. Our concert ideal sits right between them at around 1.95 milliseconds. The second lever is built into the scale design — the Young-Helmholtz law: any partial that has a vibration node right at the strike point cannot be excited. Striking at one-eighth or one-seventh of string length — 195 or 223 millimeters — deliberately suppresses harsh, rough partials like the natural seventh H7 and major second H9 while preserving the glorious singing major third H5 in full bloom!"

- **Exact Simulation Reference Values on Slide 25:**
  - `☁ Soft` (`τc ≈ 2.85 ms`): `H₆–H₁₂` drop by `−4.6 dB`; `H₁` & `H₂` sound warm and rounded.
  - `✦ Concert Ideal` (`τc ≈ 1.95 ms`): Deep foundation (`H₁–H₄`, `55–220 Hz`) blends seamlessly with singing `H₅` (major 3rd) and `H₆` (5th).
  - `⚡ Hard` (`τc ≈ 1.25 ms`): `H₆–H₁₂` surge by `+4.8 dB` (`H₁₂` up `+6.4 dB`) — bright attack, shorter sustain.
  - **Strike Points on `L = 1,560 mm`:** `1/9 L` (`173 mm`, cancels sharp `H₉ = 495 Hz`) · **`1/8 L` (`195 mm`, concert standard, node on `H₈ −22 dB`, tames `H₇` & `H₉`)** · `1/7 L` (`223 mm`, cancels natural 7th `H₇ −22 dB`).

SLIDE 26 The Hammerhead as a Complex Spring: From Cristofori to Zone Balance

- ▸ Interaction: Start on `◆ 1. Cristofori Roll (1700)` (`26.1`), then switch to `♯ 2. Renner Tension Spring` (`26.2`) and demonstrate the 3 classic workshop fault patterns (`No pp`, `No ff`, `Double Lock`) using the four 3-step sliders.

> ◈ Spoken Narrative: "To voice accurately, we must understand what a hammerhead truly is in physics: a multi-stage complex spring! Bartolomeo Cristofori solved this brilliantly back in 1700: his hammer was not a solid wooden block, but a hollow deer-leathered parchment cylinder. Soft deer leather cushioned gentle piano strokes, while the hoop tension of the hollow ring delivered 88 percent rebound and a 1.85-millisecond contact time so the hammer sprang clear of the string instantly. Now switch to our modern Renner hammerhead: compaction peaks at 100 percent right at the crown and drops to exactly half — 50 percent — across the widest flank, anchored firmly to the hammer moulding by glue and brass rivet. Watch what happens when zone balance breaks down: if the crown or transition is too hard, even the lightest touch bounces off a rigid shell — the piano loses its pianissimo! Conversely, if the flank lacks elasticity or an over-hardened shoulder blocks energy flow into the flank, contact time stretches toward 3 milliseconds — the hammer lingers on the string and chokes its own fortissimo! Remember this rule: carrying fortissimo never comes from dead surface hardness — it always comes from fast elastic rebound out of the flank!"

- **The 5 Model & Diagnostic States on Slide 26:**
  1. **Cristofori (1700):** Soft deer leather for *piano* + hollow ring tension with zero dead core mass (`88%` bounce, `τc ≈ 1.85 ms`).
  2. **Ideal Zone Balance (`pp ↔ ff`):** Supple crown entry (`100%`) flows smoothly into the elastic `½`-density flank (`50%`, `τc ≈ 1.80 ms`) → full dynamic range from *pp* to *ff*.
  3. **Fault "No Pianissimo (`pp`)":** Over-hardened crown/transition (`125%`, `+4.8 dB`, `τc ≈ 1.35 ms`) → *Remedy:* gently open the transition behind the crown with `90°` perpendicular stitches.
  4. **Fault "No Fortissimo (`ff`)":** Sluggish, inelastic felt (`54%` bounce) lingers at `τc ≈ 2.95 ms` and damps the vibrating string.
  5. **Fault "Double Lock":** Hard transition kills *pp* while decoupling the flank (`18%` active, `τc → 2.75 ms`) → neither *pp* nor *ff*!

SLIDE 27 Voicing Zones & Needling Effect: The Renner Clock Model

- ▸ Interaction: First demonstrate the technique toggle on the right (`2 · ✓ ⊥ to Fiber` vs. `3 · × Cross / Chop`); then click through all **6 color zones** (`Blue`, `Green`, `Orange`, `Purple`, `Yellow Rivet→Flank`, `Red 12:00 Taboo`) and audition each with `Needle & Play`.

> ◈ Spoken Narrative: "This brings us to the heart of practical needle work: the Renner Clock Model, rooted in the methodology of our long-standing master voicer Burkard Olbrich. Before touching a single zone, one iron law governs everything: always follow the fiber course of the felt wrapped around the moulding! Across the entire upper arc from 9 o'clock over 12 o'clock to 3 o'clock, always press the needle in smoothly at right angles — 90 degrees perpendicular to the fiber course — never chopping and never angling across the grain! Why? Because we want to part those parallel-carded fibers gently without severing a single tensioned outer fleece web. Let us walk from outside inward through all six zones — and pay special attention to our yellow tension-building stitch just above the rivet!"

- **All 6 Zones with Tooling, Depth & Measured Acoustic Effect:**
  1. ● **Blue `ff`-Wedge (`9–10` & `2–3 o'clock` — Pre-Voicing):** 3-needle holder · `7–9 mm` **deep** (`⊥` to fiber, stopping just short of the underfelt!) → relieves press hardness, channels elasticity upward into the `f`-wedge, boosts fundamental power (`H₁–H₄ +3.2 dB`), and unlocks free *fortissimo*.
  2. ● **Green `f`-Wedge (`10–11` & `1–2 o'clock` — Mid-Voicing):** `4–6 mm` **deep** (`⊥` to fiber) → opens the shoulder bridge, lets the flank flex freely, shapes the singing cantabile core (`H₃–H₆ +2.4 dB`), and trims harsh glare (`–2.5 dB`).
  3. ● **Orange `p`-Wedge (`11–12` & `12–1 o'clock` — Balancing):** Fine needle · `2.5–4 mm` **deep** (`⊥` to fiber) → cushions glassy upper partials (`H₈–H₁₂ –3.8 dB`) for a warm, rounded *piano* while keeping full *forte* support.
  4. ● **Purple `pp`-Wedge (Crown Surface — Fine Voicing):** Short single needle · `max. 1–2 mm` (`≤ 2.0 mm`) `⊥` to fiber and **strictly between the string grooves!** → creates a velvety micro-cushion (`H₉–H₁₂ –5.1 dB`) while keeping `96%` core tension intact.
  5. ● **Tension Stitch: Above Rivet → Angled Steeply Toward Flank: Deliberately few, deep stitches (`10–14 mm`)** entering just above the rivet and angled steeply upward into the lower flank → tightens lower flank support, shortens contact time (`τc ≈ 1.58 ms`), raises elasticity, and **actively boosts upper partials (`H₇–H₁₂ +3.1 dB`)!**
  6. ● **Red Taboo Zone (`12:00` Deep Crown Core): Strict Taboo (`> 2 mm` forbidden)!** → Severs the outer tension arch (`σ` collapses to `34%`, `τc` jumps to `3.15 ms`, `–6.8 dB`) — leaving the tone permanently hollow, woolly, and dead!

SLIDE 28 Crown & String Plane in 3-String Choirs: Phase & Cancellation

- ▸ Interaction: Cycle through all **4 alignment buttons** (`✓ Flat & Parallel` · `∠ Tilted Crown` · `↕ Uneven String Plane` · `⌣ Hollow Crown`), adjust the angle/height sliders, and click `Play A₃ Choir` (`220 Hz`) to make comb-filter cancellation immediately audible!

> ◈ Spoken Narrative: "In everyday workshop life, many technicians reach for the voicing needle far too early whenever a treble or midrange note buzzes, sounds metallic, or feels hollow. Slide 28 reveals the real culprit on a three-string A3 choir at 220 Hertz: phase cancellation! Whenever a hammerhead sits slightly tilted, a single wire at the agraffe or bridge rides just two or three tenths of a millimeter out of plane, or deep grooves wear a hollow crown, the three strings S1, S2, and S3 are struck fractions of a millisecond apart. For the long wavelength of the fundamental that barely matters — yet for the short wavelengths of mid and upper partials, that tiny arrival delay suddenly equals a 120- to 180-degree phase shift! Wave crest meets wave trough, and the three strings cancel one another out in deep comb-filter notches. Therefore, before touching any needle: bring hammer travel, crown flatness, and the 3-string plane into 100 percent alignment first!"

- **The 4 Alignment States on Slide 28:**
  1. `✓ Flat & Parallel` (`Δt = 0 ms`, `Δφ = 0°`): Constructive `3×` summed amplitude with zero comb-filter notches.
  2. `∠ Tilted Crown`: Staggered impact `S₁ → S₂ → S₃` → metallic buzz & upper-partial cancellation (`Δφ = 120°–240°`).
  3. `↕ Uneven String Plane` (`0.2–0.45 mm`): One proud or low wire fights the choir at `Δφ ≈ 180°` → hollow tone & false beats despite pure tuning.
  4. `⌣ Hollow Crown / Grooves`: Outer wires `S₁/S₃` strike early while center wire `S₂` drops out in *pp* and cancels out of phase in *f* → resurface cleanly with `320–400 grit` first!

SLIDE 29 String Preparation: Hitch Pins, Bridge Pins & Agraffes

> **◈ Spoken Narrative:** “Directly from that phase physics comes our mandatory four-step string checklist — photographed right during our Yamaha C3 kit installation in Meuselwitz — before picking up a single voicing needle: First, we clean and inspect solid string seating around the hitch pins. Second, we tap every string firmly down onto the bridge pins using a brass drift punch and light hammer to eliminate false beats. Third, we space and level all three choir wires behind the agraffes with fine pliers into one exact plane. And fourth, we verify clean damper seating, beatless unison tuning, and simultaneous hammer-to-string contact.”

- **The 4 Mandatory Pre-Voicing Steps:**
  1. **01 · Hitch Pins & Bearing:** Clean bearing surfaces and verify solid, stress-free seating.
  2. **02 · Bridge Pins:** Seat strings cleanly onto the bridge with a brass drift punch (eliminates false beats).
  3. **03 · Agraffes & String Plane:** Bring all 3 unison wires to equal lateral spacing and identical height.
  4. **04 · Choir & Damper Check:** Simultaneous hammer contact across all 3 strings + 100% pure unison tuning.

SLIDE 30 Intermediate, Fine & Una Corda Voicing

> **◈ Spoken Narrative:** “Once action geometry, crown, and string plane are spot-on, five refined steps bring all 88 notes to true concert maturity: We work with a short single needle in the ergonomic BDK Fine Holder, first evening out any shoulder transitions on a firm support block, and then performing fine crown voicing right inside the grand through the strings using the Chop-Stick — maximum two millimeters deep and strictly between the string grooves! Next, we smooth raised surface fibers with a fine 320- to 400-grit sanding paddle. And never skip Step 5: shift-pedal voicing for una corda! The moment the pianist depresses the left pedal, the strings shift onto the unplayed ridges between the grooves — if those ridges bark brightly or metallically, even the finest open voicing is spoiled. So depress the soft pedal, delicately needle only the newly contacted ridge zones, and lightly chamfer the left crown edge.”

- **The 5 Steps Including Tool Part Numbers:**
  1. **BDK Fine Holder (`Part No. 43-0003269`):** Short single needle for pinpoint tactile control.
  2. **Intermediate Voicing (Shoulder Zone):** Chromatic balancing across the forte and piano wedges on a solid support pad.
  3. **2 mm Crown Stitches Inside the Grand:** Reach between the choir wires with the **Chop-Stick (`Part No. 43-0003271`) — max. `2 mm` deep strictly between string grooves!**
  4. **Fine Crown Smoothing:** Smooth raised wool fibers with a fine sanding file (`320–400 grit`).
  5. **Shift-Pedal Voicing (*Una Corda*):** With soft pedal depressed, equalize newly struck ridge zones & lightly chamfer the left crown edge.

SLIDE 31 Lacquer Dosing, Heat Ironing (max. 135 °C), Crown Finish & Weight

- ▸ Interaction: Click all **4 lacquer zones** on the left (● Rivet Zone · ● Lateral Above Moulding · ⚠ Lateral: Excess Lacquer · ● Direct on Crown), toggle the time-compaction button (⏱ **Freshly Applied** ↔ Aged compaction over play), and click the mass buttons (**+m ↑** vs. **−m ↓**) at the bottom.

> ◈ **Spoken Narrative:** "What if a hammerhead needs to become brighter, tauter, or more projecting rather than softer? Slide 31 shows all four tools — and above all their limits! Let's start on the left with lacquer-based hardeners: dosing down at the rivet zone — below the hammer moulding — stiffens the lower foundation by plus 3.4 decibels and, depending on lacquer shrinkage as it cures, can gently lift outer wrap tension without touching the elastic impact arch above. By contrast, applying diluted lacquer laterally above the hammer moulding boosts high partials H9 through H12 — but watch what happens when we click the time-factor toggle: every lacquer application above the moulding continues to compact further under concert play over time! Always leave a safety margin — and never apply excess lacquer, or it wicks into the flank and locks up the spring completely. And putting hardener directly onto the 12 o'clock crown is an absolute taboo: it forms a glassy, nasal crust and ruins the hammerhead permanently! Whenever an instrument lacks true fundamental fullness and projection, never chase it with excess lacquer — solve it via mass: from the middle register into the bass, slightly increase hammer weight; moving toward the treble, reduce hammer weight! On the right you see the two safest non-chemical alternatives: First, heat ironing with a thermostat-controlled voicing iron strictly capped at 135 degrees Celsius — hotter than that scorches keratin and yellows the lanolin. And second, our insider workshop tip for ultra-fine crown finishing: polishing with micro-abrasive film down to 15 microns — or trimming stray damping fuzz fibers with an ordinary Braun Series 1 foil shaver at zero percent felt removal!"

- **Compact Workshop Summary for Slide 31:**
  - **01 · Lacquer Zones:** ● Rivet (H_1–H_4 +3.4 dB, safe below moulding) · ● Lateral above moulding (H_9–H_{12} +3.8 dB, **compacts further over playing time!**) · ⚠ Excess lateral (glues flank shut, H_1–H_4 −3.8 dB) · ● Crown 12:00 (**Taboo!**, nasal +6.0 dB, permanent damage).
  - **Mass Control Instead of Over-Hardening:** Middle → Bass **+m ↑** (greater impact energy & fundamental) · Toward Treble **−m ↓** (faster acceleration, shorter τc).
  - **02 · Heat Ironing:** Mandatory thermostat **strictly max.** **135 °C**, brisk strokes following the crown contour.
  - **03 · Ultra-Fine Crown Finish:** Micro-abrasive film **≤ 15 µm** **(15 my)** for mirror-smooth string contact or **Braun Series 1** foil shaver (0% felt loss, 100% spring preservation).

SLIDE 32 The Renner Academy & Technical Master Dialogue

> ◈ **Spoken Narrative:** "Everything we have explored together today in physics, laboratory data, and interactive simulation truly comes alive the moment you hold needle, shank, and gauge in your own hands. That is why I warmly invite you to the Renner Academy and to our newly unified headquarters in Meuselwitz: under our motto 'By Masters for Masters,' we host hands-on intensive workshops covering action geometry, hammerhead installation, Yamaha Kit conversion, and concert voicing — paired with a live tour through every production department and direct technical support for your workshop."

SLIDE 33 Let the Perfect Hammer Sound!

> ◈ **Closing Words & Opening the Floor for Q&A:** "From intact Merino wool fiber beside the Mulde river in Wurzen to a hammerhead machined to one-tenth of a millimeter and voiced to concert perfection in Meuselwitz, we have pursued a single mission since 1882: placing every imaginable shade of tonal color beneath the pianist's fingers — Let the Perfect Hammer Sound! Thank you very much for your attention, and I now look forward to your questions, your workshop experiences, and an open technical dialogue!"

Appendix: Master-Level Answers to Frequent Conference Questions (Q&A Reference)

Q&A GUIDE Workshop Q&A — Top 6 Technician Questions at Symposia & Masterclasses

- **Question 1: "Why does FFW Wurzen stop fulling at 56% (0.56 g/cm³) — aren't harder felts available on the market?"**
  - **Answer:** 56% is the natural physical ceiling of purely mechanical felting in pure Merino wool. Higher apparent hardness can only be forced by excessive mechanical pressing (which crushes hollow keratin fibers — **felt breakage**, killing elastic rebound) or by adding synthetic resins/chemical hardeners, which increase internal damping friction, narrow the overtone spectrum, and react unpredictably to humidity swings.

- **Question 2: "How does Weickert Special perform under extreme Asian humidity swings (humid summer/monsoon vs. dry heated winter)?"**
  - **Answer:** That is precisely what accredited **STFI Test Report P2024 0903** quantified: thanks to `+38%` **higher natural residual wool grease (`0.36%` lanolin per ISO 3074)**, the wool fiber absorbs dramatically less ambient moisture — thickness swelling during the 48-hour climate cycling test drops by `65%` (`4.81%` vs. `13.69%`). Voicing and regulation remain vastly more climate-stable.
- **Question 3: "Which exact models fit the Louis Renner Yamaha Kit, and does it still require voicing after installation?"**
  - **Answer:** The kit is custom-engineered for **Yamaha C1, C2, and C3** grand pianos. Thanks to five years of developing our proprietary measurement and calibration system, the set leaves Meuselwitz already calibrated to a warm European concert spectrum **requiring zero manual pre-voicing** — you simply align hammers to strings, regulate the action (`4–5 h` on site), and optionally tailor final micro-nuances to the room acoustics.
- **Question 4: "When do you recommend hornbeam versus American walnut for the hammer moulding?"**
  - **Answer: Hornbeam** (when gently and patiently seasoned) is our ideal, heavy, remarkably uniform all-round moulding wood delivering strong foundational projection. **American walnut** is our lightest, tightest-grained, longest-fibered premier tonewood — unbeatable wherever low rotational inertia, lightning-fast repetition, and a radiant, non-self-damping treble are paramount (including our Yamaha Kit).
- **Question 5: "Why must 12 o'clock crown stitches stay under 2 mm deep and strictly between the string grooves?"**
  - **Answer:** The outermost `2 × 6` unsanded fleece webs form the continuous tensile arch carrying the hammerhead's rebound force. Piercing deeper than `2 mm` into the red `12:00` core severs that arch irreversibly (`τc` lengthens sharply and the tone turns hollow and dead in both *p* and *ff*). Needling directly inside the string groove tears up the exact contact patch and introduces uneven phase errors across the unison choir.
- **Question 6: "Why use a Braun Series 1 foil shaver on the crown instead of ordinary sandpaper?"**
  - **Answer:** Coarse sanding strips away precious outer tension-web material, altering both geometry and mass. Micro-abrasive film (`≤ 15 µm`) polishes only the contact apex, while the **Braun Series 1** foil screen clips solely stray protruding fuzz fibers that blur the initial transient — achieving `0%` **felt loss** and 100% preservation of the tension spring.